

Younes Ben Mazziane

Research Scientist — Algorithms, AI/ML, and Network Optimization
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SUMMARY

Ph.D. computer scientist specializing in the **mathematical performance analysis of online algorithms**, with applications to caching and hash-based network telemetry. Expertise in **probability theory, online learning, and stochastic optimization**, complemented by Python-based simulation and validation on synthetic and real-world traces. Published in **ACM SIGMETRICS/POMACS** and **IEEE/ACM Transactions on Networking**.

TECHNICAL SKILLS

- **Programming skills:** Python, Pytorch, SciPy, scikit-learn, stochastic simulation, data analysis, Git.
- **Methods:** Probability theory, Markov chains, mean-field approximations, online learning, stochastic optimization, stochastic approximation, game theory, dynamical systems.
- **Applications:** Network telemetry and measurement, caching, similarity-based retrieval systems, sketching algorithms, resource allocation systems.

EXPERIENCE

Postdoctoral Researcher
Avignon University

Nov. 2024 – Oct. 2026
Avignon, France

- Derived mathematical **closed-form performance characterizations** for Elastic Sketch, a modern hash-based network telemetry algorithm, enabling efficient performance prediction and used **stochastic optimization** to derive workload-aware configuration strategies that improve its practical deployment.
- Studied **distributed online learning** in multi-agent resource-allocation systems, including auction- and price-based mechanisms, using **game theory** and **dynamical-systems** techniques, and established convergence to Nash equilibria and social-welfare guarantees.
- Theoretically established limitations of empirically effective **online-learning-based caching policies** for combining LRU and LFU (e.g., LeCaR) under adversarial request sequences, and designed **efficient and robust no-regret alternative** with provable performance guarantees.
- Collaborated with researchers from **Meta, Inria, IIT Bombay, UFRJ, and Linköping University**.
- Developed Python simulation frameworks with GUI and command-line interfaces (Goody, argparse) to validate analytical predictions through **Monte Carlo simulations** on synthetic and real-world traces, using **NumPy, SciPy, scikit-learn, PyTorch, and Matplotlib**.
- Published research in **ACM SIGMETRICS, IFIP Performance, IEEE WiOpt, and IEEE/ACM Transactions on Networking**.

Doctoral Researcher
Inria and Université Côte d'Azur

Oct. 2020 – Oct. 2024
Sophia Antipolis, France

- Mathematically quantified the performance of widely used randomized algorithms under stochastic input models using **probability theory, Markov chains, and mean-field approximations**, including:
 - * Count-Min Sketch with conservative updates, a hash-based algorithm for **network telemetry and anomaly detection**.
 - * Extensions of Least-Recently-Used (LRU) caching for **similarity-based retrieval systems**.
- Designed **online-learning and no-regret caching algorithms** that, for arbitrary request sequences, asymptotically outperform the optimal static policy, even under partial observability of the requests, while requiring only $O(1)$ expected amortized update time.
- Developed Python simulation frameworks with GUI and command-line interfaces (Goody, argparse) to validate analytical predictions through **Monte Carlo simulations** on synthetic and real-world traces, using **NumPy, SciPy, scikit-learn, PyTorch, and Matplotlib** (see for example [BPO caching github])
- Published research in **ACM SIGMETRICS, IEEE/ACM Transactions on Networking, Computer Networks, IEEE GLOBECOM, and IEEE INFOCOM Workshops**.

Teaching Assistant — Python Programming
Université Côte d'Azur / Polytech Nice Sophia

Sep. 2023 – Aug. 2024
Sophia Antipolis, France

- Taught 80 hours of introductory and advanced Python programming through practical sessions and exercises.

EDUCATION

Ph.D. in Computer Science
Université Côte d'Azur — Thesis: Probabilistic Analysis for Caching

Oct. 2020 – Oct. 2024
Sophia Antipolis, France

M.Sc. in Computer Science — Ubiquitous Networking
Polytech Nice Sophia, Université Côte d'Azur

Sep. 2019 – Aug. 2020
Sophia Antipolis, France

Engineering Degree / M.Sc. in Electrical and Computer Engineering
Institut National des Postes et Télécommunications

Sep. 2017 – Aug. 2020
Rabat, Morocco

SELECTED PUBLICATIONS

- [SIGMETRICS] **Younes Ben Mazziane**, Vinay Kumar B. R., Othmane Marfoq
Elastic Sketch under Random Stationary Streams: Limiting Behavior and Near-Optimal Configuration
ACM SIGMETRICS / POMACS, 2027
- [TON] **Younes Ben Mazziane**, Francescomaria Faticanti, Sara Alouf, Giovanni Neglia
Efficient and Optimal No-Regret Caching under Partial Observation
IEEE/ACM Transactions on Networking, 2026
- [SIGMETRICS] **Younes Ben Mazziane**, Othmane Marfoq
Universal and Tight Bounds on Counting Errors of Count-Min Sketch with Conservative Updates
ACM SIGMETRICS / POMACS, 2025

PUBLICATIONS

2027

- [SIGMETRICS] **Younes Ben Mazziane**, Vinay Kumar B. R., Othmane Marfoq
Elastic Sketch under Random Stationary Streams: Limiting Behavior and Near-Optimal Configuration
ACM SIGMETRICS / POMACS

2026

- [Preprint] **Younes Ben Mazziane**, Cleque-Marlain Mboulou-Moutoubi, Francesco De Pellegrini, Eitan Altman
Learning in Proportional Allocation Games
Preprint – Major Revision at IEEE/ACM Transactions on Networking
- [TON] **Younes Ben Mazziane**, Francescomaria Faticanti, Sara Alouf, Giovanni Neglia
Efficient and Optimal No-Regret Caching under Partial Observation
IEEE/ACM Transactions on Networking
- [Performance] **Younes Ben Mazziane**, Xinying Zou
No-Regret Mixing of LRU and LFU with Optimal Switching Cost
IFIP Performance
- [WiOpt-W] Cleque-Marlain Mboulou-Moutoubi, **Younes Ben Mazziane**, Eitan Altman, Francesco De Pellegrini
Repeated Multi-Resource Proportional Allocation Auction Games
IEEE WiOpt – RAWNET Workshop

2025

- [SIGMETRICS] **Younes Ben Mazziane**, Othmane Marfoq
Universal and Tight Bounds on Counting Errors of Count-Min Sketch with Conservative Updates
ACM SIGMETRICS / POMACS
- [NETGCOOP] Cleque-Marlain Mboulou-Moutoubi, **Younes Ben Mazziane**, Francesco De Pellegrini, Eitan Altman
Best-Response Learning in Budgeted α -Fair Kelly Mechanisms
NETGCOOP
- [WiOpt] **Younes Ben Mazziane**, Cleque-Marlain Mboulou-Moutoubi, Francesco De Pellegrini, Eitan Altman
Learning to Bid in Proportional Allocation Auctions with Budget Constraints
IEEE WiOpt

2024

- [ComNet] **Younes Ben Mazziane**, Sara Alouf, Giovanni Neglia, Daniel S. Menasché
TTL Model for an LRU-based Similarity Caching Policy
Elsevier Computer Networks

2023

- [VCC] **Younes Ben Mazziane**, Francescomaria Faticanti, Giovanni Neglia, Sara Alouf
No-Regret Caching with Noisy Request Estimates
IEEE VCC

2022

- [ComNet] **Younes Ben Mazziane**, Sara Alouf, Giovanni Neglia
Analyzing Count-Min Sketch with Conservative Updates
Elsvier Computer Networks
- [GLOBECOM] **Younes Ben Mazziane**, Sara Alouf, Giovanni Neglia, Daniel S. Menasché
Computing the Hit Rate of Similarity Caching
IEEE GLOBECOM
- [INFOCOM-W] **Younes Ben Mazziane**, Sara Alouf, Giovanni Neglia
A Formal Analysis of the Count-Min Sketch with Conservative Updates
IEEE INFOCOM Workshops

SOFTWARE

- [BPO caching] This repository provides a simulation tool for comparing caching policies on synthetic request sequences while allowing for partial observation of the requests. The considered caching policies are LRU, LFU, and variants of Follow-The-Perturbed-Leader algorithm in caching.

LANGUAGES

Languages: English, French, Arabic — full professional proficiency.

ACADEMIC SERVICE

2027

- Technical Program Committee member, IEEE INFOCOM.
- Technical Program Committee member, ACM SIGMETRICS.

2026

- Shadow Program Committee member, ACM SIGMETRICS.
- Reviewer, IEEE INFOCOM.

Journal Reviewing

- Elsevier Performance Evaluation.
- Elsevier Computer Communications.
- IEEE Networking Letters.